

DSA0603 — DATA HANDLING AND VISUALIZATION

CO3_AT2 — DATA ANALYSIS PROJECT

Course Outcome

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods.

QUESTION 1 — Comparing Distributions with ECDF and Q-Q Plots

1. Objective

The objective is to compare the delivery-time distributions of **Partner A** and **Partner B** using ECDF and normal Q-Q plots. The analysis determines which courier is consistently faster, whether the delivery times are approximately normally distributed, and which method should be used for setting operational SLA limits.

2. R Code

```
# =====  
  
# QUESTION 1 - ECDF AND Q-Q PLOTS  
  
# =====  
  
# Install packages if required  
# install.packages("ggplot2")  
# install.packages("gridExtra")  
  
library(ggplot2)  
library(gridExtra)  
  
# -----  
# Create the dataset  
# -----
```

```
partner_A <- c(18, 19, 20, 20, 21, 21, 22, 22, 23, 23,  
              24, 24, 25, 26, 27, 28, 30, 33, 38, 46)
```

```
partner_B <- c(20, 21, 22, 22, 23, 23, 23, 24, 24, 24,  
              24, 25, 25, 25, 26, 26, 27, 27, 28, 29)
```

```
delivery_data <- data.frame(  
  Time = c(partner_A, partner_B),  
  Partner = factor(  
    c(rep("Partner A", 20), rep("Partner B", 20))  
  )  
)
```

```
# -----  
# Summary statistics  
# -----
```

```
summary_A <- c(  
  Mean = mean(partner_A),  
  SD = sd(partner_A),  
  Median = median(partner_A),  
  Q1 = quantile(partner_A, 0.25),  
  Q3 = quantile(partner_A, 0.75)  
)
```

```
summary_B <- c(  
  Mean = mean(partner_B),  
  SD = sd(partner_B),
```

```

Median = median(partner_B),
Q1 = quantile(partner_B, 0.25),
Q3 = quantile(partner_B, 0.75)
)

print("Partner A Summary:")
print(summary_A)

print("Partner B Summary:")
print(summary_B)

# =====
# TASK 1: OVERLAID ECDF
# =====

ecdf_plot <- ggplot(delivery_data,
                    aes(x = Time, color = Partner)) +
  stat_ecdf(linewidth = 1.2) +
  labs(
    title = "Empirical CDF of Delivery Times",
    x = "Delivery Time (hours)",
    y = "Cumulative Proportion"
  ) +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "bottom"
  )

```

```
)
```

```
print(ecdf_plot)
```

```
# =====
```

```
# TASK 2: NORMAL Q-Q PLOTS
```

```
# =====
```

```
qq_A <- ggplot(data.frame(Time = partner_A),
```

```
  aes(sample = Time)) +
```

```
  stat_qq() +
```

```
  stat_qq_line(color = "red", linewidth = 1) +
```

```
  labs(
```

```
    title = "Normal Q-Q Plot - Partner A",
```

```
    x = "Theoretical Quantiles",
```

```
    y = "Sample Quantiles"
```

```
  ) +
```

```
  theme_minimal()
```

```
qq_B <- ggplot(data.frame(Time = partner_B),
```

```
  aes(sample = Time)) +
```

```
  stat_qq() +
```

```
  stat_qq_line(color = "red", linewidth = 1) +
```

```
  labs(
```

```
    title = "Normal Q-Q Plot - Partner B",
```

```
    x = "Theoretical Quantiles",
```

```
    y = "Sample Quantiles"
```

```
  ) +
```

```
theme_minimal()
```

```
grid.arrange(qq_A, qq_B, ncol = 2)
```

```
# =====
```

```
# TASK 3: ECDF MAXIMUM DIVERGENCE
```

```
# =====
```

```
# Calculate ECDF values at all observed time points
```

```
all_times <- sort(unique(c(partner_A, partner_B)))
```

```
ecdf_A <- ecdf(partner_A)
```

```
ecdf_B <- ecdf(partner_B)
```

```
ecdf_comparison <- data.frame(
```

```
  Time = all_times,
```

```
  Partner_A_ECDF = ecdf_A(all_times),
```

```
  Partner_B_ECDF = ecdf_B(all_times)
```

```
)
```

```
ecdf_comparison$Absolute_Difference <-
```

```
  abs(ecdf_comparison$Partner_A_ECDF -
```

```
    ecdf_comparison$Partner_B_ECDF)
```

```
print(ecdf_comparison)
```

```
max_difference_row <-
```

```
ecdf_comparison[  
  which.max(ecdf_comparison$Absolute_Difference),  
]
```

```
print("Maximum ECDF divergence:")  
print(max_difference_row)
```

```
# =====  
# TASK 4: SKEWNESS AND OUTLIER CHECK  
# =====
```

```
# install.packages("moments")  
library(moments)
```

```
cat("Partner A skewness:",  
    skewness(partner_A), "\n")
```

```
cat("Partner B skewness:",  
    skewness(partner_B), "\n")
```

```
# =====  
# TASK 5: EMPIRICAL SLA PERCENTILES  
# =====
```

```
sla_A <- quantile(partner_A, c(0.90, 0.95, 0.99))  
sla_B <- quantile(partner_B, c(0.90, 0.95, 0.99))
```

```
print("Partner A SLA percentiles:")
```

```
print(sla_A)
```

```
print("Partner B SLA percentiles:")
```

```
print(sla_B)
```

3. ECDF Interpretation

The ECDF represents the proportion of shipments completed within each delivery-time threshold.

The maximum difference occurs at approximately **29 hours**.

At 29 hours:

- Partner A has completed approximately **80%** of shipments.
- Partner B has completed **100%** of shipments.
- Difference = **20 percentage points**.

Therefore, the maximum ECDF divergence is:

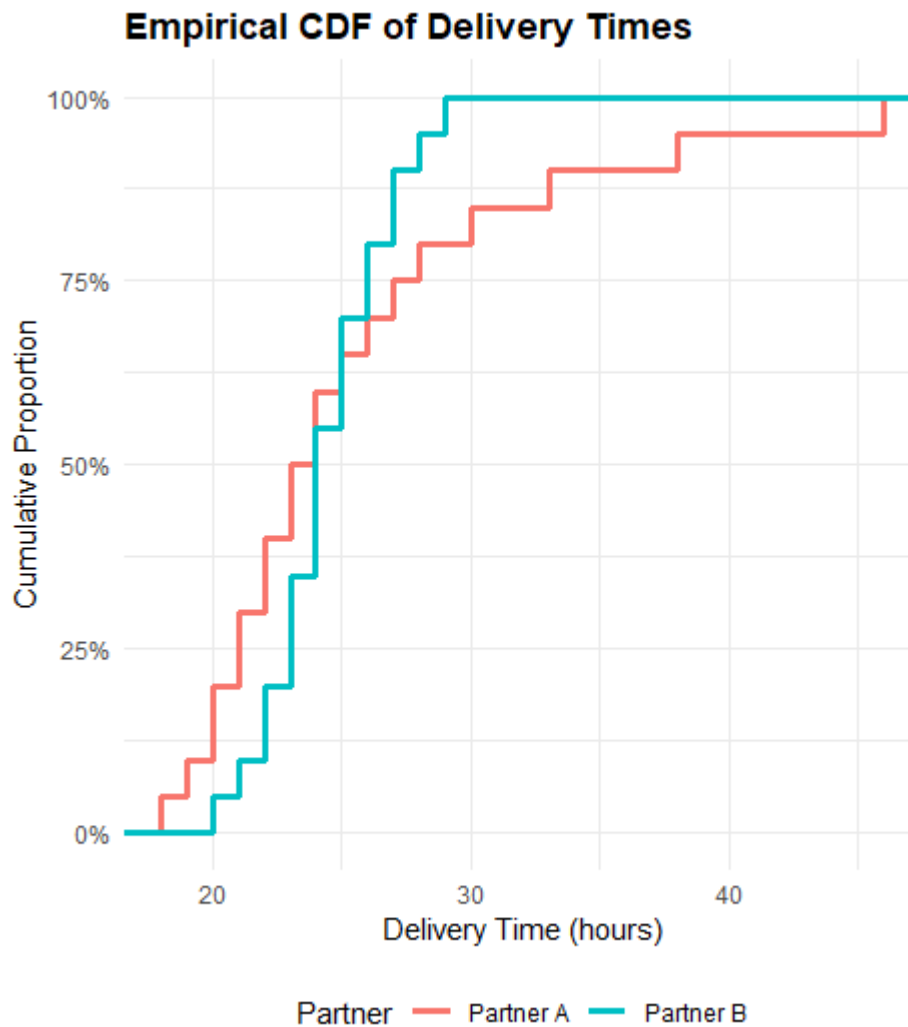
```
[  
D = |0.80 - 1.00| = 0.20  
]
```

Operational interpretation

At the **29-hour threshold**, Partner B has delivered all of its observed shipments, whereas Partner A has delivered only 80%.

This indicates that Partner A has a **longer upper tail**. The observations of 30, 33, 38 and 46 hours are responsible for the delayed deliveries.

Partner B is therefore more consistent for avoiding very long delivery times.



4. Q-Q Plot Interpretation

Partner A

Partner A does **not** strongly support the normal-distribution assumption.

The Q-Q plot is expected to show noticeable deviation from the straight reference line, especially at the upper tail.

The value of **46 hours** is particularly extreme compared with the rest of the observations.

Partner A therefore shows:

- Positive/right skewness
- A long upper tail
- Possible extreme observation at 46 hours

- Greater variability

Mean:

```
[
\bar{x}_A = 25.5
]
```

Standard deviation:

```
[
s_A \approx 6.89
]
```

Median:

```
[
Median_A = 23.5
]
```

The mean being considerably larger than the median is additional evidence of right skew.

Partner B

Partner B is considerably closer to a normal-looking distribution.

Its observations are concentrated between approximately 20 and 29 hours and do not contain an extreme observation comparable to Partner A's 46-hour value.

Mean:

```
[
\bar{x}_B = 24.4
]
```

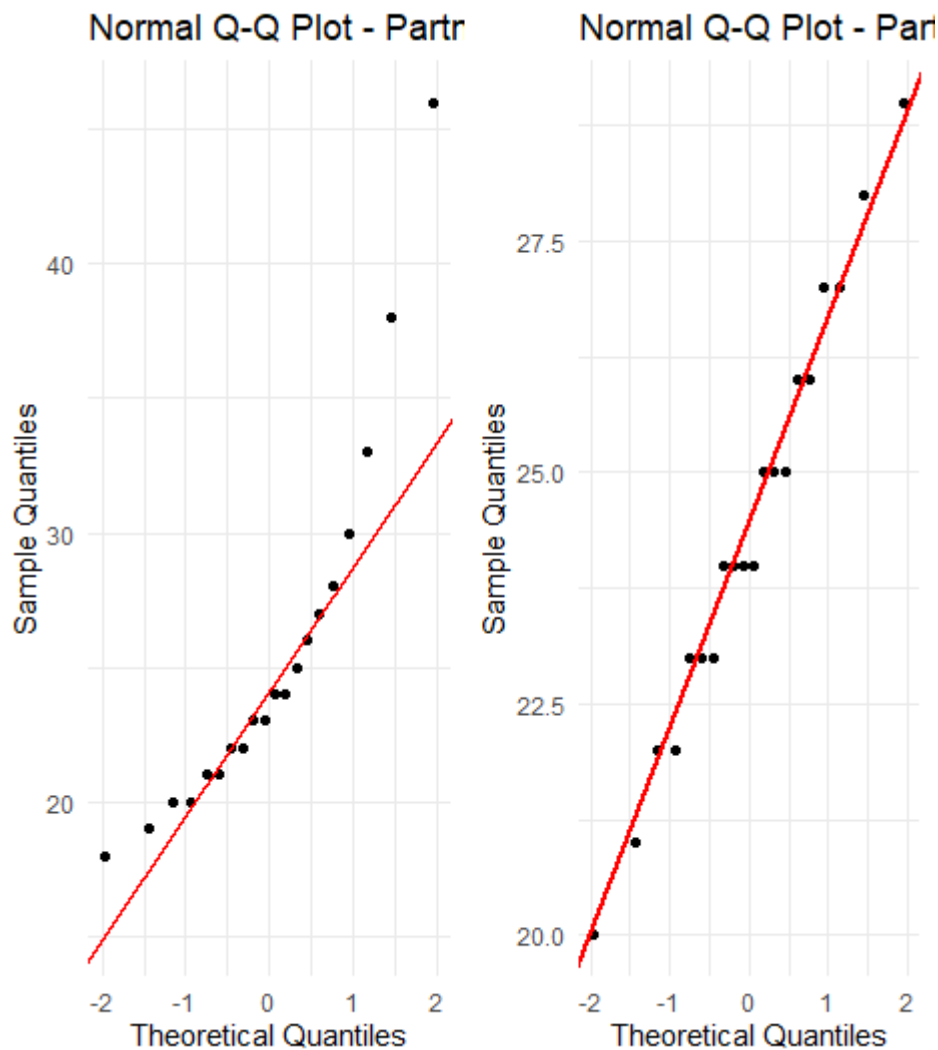
Standard deviation:

```
[
s_B \approx 2.33
]
```

Median:

```
[
Median_B = 24
]
```

Partner B therefore has substantially lower variability.



5. SLA Recommendation

The recommended approach is to use **empirical percentiles rather than mean $\pm$ standard deviation**.

The reason is that Partner A is right-skewed and contains an extreme upper-tail observation.

Mean $\pm$ standard deviation assumes that the mean and standard deviation provide a reliable representation of the distribution. This is less appropriate when the distribution is skewed.

For operational SLAs, a percentile such as the **90th or 95th percentile** directly represents the observed delivery-time performance.

For example, the approximate 95th percentiles are:

- Partner A: **38.4 hours**
- Partner B: **28.05 hours**

Therefore, percentile-based SLA limits are more robust and operationally meaningful.

Final conclusion

Partner B is the more consistent courier partner. Partner A has a longer right tail and greater variability. Empirical percentile-based SLAs are recommended.

QUESTION 2 — Distribution Shape Across Many Groups Using Bean Plots

1. Objective

The objective is to compare waiting-time distributions across five hospital departments and identify variability, multimodality and departments requiring staffing intervention.

2. R Code

```
# =====

# QUESTION 2 - BEAN / DENSITY + JITTER PLOTS

# =====

library(ggplot2)

# -----
# Create data
# -----

waiting_data <- data.frame(
  Department = c(
    rep("Cardiology", 10),
    rep("Orthopedics", 10),
    rep("Pediatrics", 10),
    rep("Dermatology", 10),
```

```
rep("General Medicine", 10)
),
```

```
Waiting_Time = c(
  10, 12, 13, 14, 15, 15, 16, 17, 18, 20,
  8, 9, 25, 26, 27, 9, 10, 28, 29, 8,
  5, 6, 6, 7, 7, 8, 8, 9, 9, 10,
  15, 16, 17, 18, 19, 20, 21, 22, 23, 24,
  10, 40, 12, 38, 11, 41, 13, 39, 12, 40
)
)
```

```
# -----
# Summary statistics
# -----
```

```
summary_table <- aggregate(
  Waiting_Time ~ Department,
  data = waiting_data,
  FUN = function(x) {
    c(
      Mean = mean(x),
      Median = median(x),
      SD = sd(x),
      Minimum = min(x),
      Maximum = max(x),
      Range = max(x) - min(x)
    )
  }
)
```

```

    }
  )

print(summary_table)

# =====

# TASK 1: BEAN-LIKE VISUALIZATION

# =====

bean_plot <- ggplot(
  waiting_data,
  aes(
    x = Waiting_Time,
    y = Department
  )
) +
  geom_density_ridges(
    aes(fill = Department),
    alpha = 0.5,
    scale = 0.8,
    rel_min_height = 0.01
  ) +
  geom_jitter(
    width = 0,
    height = 0.08,
    size = 2,
    alpha = 0.7
  ) +

```

```

labs(
  title = "Waiting-Time Distributions by Department",
  x = "Waiting Time (minutes)",
  y = "Department"
) +
theme_minimal() +
theme(
  legend.position = "none",
  plot.title = element_text(face = "bold")
)

print(bean_plot)

# =====

# ALTERNATIVE BEAN PLOT WITHOUT GGRIDGES

# =====

density_jitter <- ggplot(
  waiting_data,
  aes(x = Waiting_Time, y = Department)
) +
  geom_violin(
    fill = "lightgray",
    alpha = 0.7,
    trim = FALSE
  ) +
  geom_jitter(
    width = 0,

```

```

    height = 0.08,
    size = 2
  ) +
  labs(
    title = "Density-plus-Jitter Visualization of Waiting Times",
    x = "Waiting Time (minutes)",
    y = "Department"
  ) +
  theme_minimal()

print(density_jitter)

# =====

# TASK 3: STANDARD BOXPLOT

# =====

box_plot <- ggplot(
  waiting_data,
  aes(
    x = Waiting_Time,
    y = Department
  )
) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Waiting Times",
    x = "Waiting Time (minutes)",
    y = "Department"
  )

```

```
) +
```

```
theme_minimal()
```

```
print(box_plot)
```

```
# =====
```

```
# TASK 4: RANK DEPARTMENTS
```

```
# =====
```

```
median_ranking <- aggregate(
```

```
  Waiting_Time ~ Department,
```

```
  data = waiting_data,
```

```
  FUN = median
```

```
)
```

```
median_ranking <- median_ranking[
```

```
  order(-median_ranking$Waiting_Time),
```

```
]
```

```
print("Ranking by Median Waiting Time:")
```

```
print(median_ranking)
```

```
sd_ranking <- aggregate(
```

```
  Waiting_Time ~ Department,
```

```
  data = waiting_data,
```

```
  FUN = sd
```

```
)
```



```
sd_ranking <- sd_ranking[  
  order(-sd_ranking$Waiting_Time),  
]
```

```
print("Ranking by Variability:")
```

```
print(sd_ranking)
```

If `geom_density_ridges()` gives an error, install the required package first:

```
install.packages("ggridges")
```

and add:

```
library(ggridges)
```

3. Distribution Analysis

Cardiology

Cardiology has waiting times between **10 and 20 minutes**.

The values are relatively concentrated around 15 minutes.

Therefore, Cardiology has **moderate and predictable variability**.

Orthopedics

Orthopedics contains two clearly separated groups:

- 8–10 minutes
- 25–29 minutes

This creates a **bimodal distribution**.

This could indicate two different patient groups, appointment types, staffing conditions, or different service processes.

Pediatrics

Pediatrics has the narrowest distribution, from **5 to 10 minutes**.

It is highly consistent and has low variability.

Dermatology

Dermatology has a relatively smooth distribution from **15 to 24 minutes**.

It has moderate variability without an obvious strong second mode.

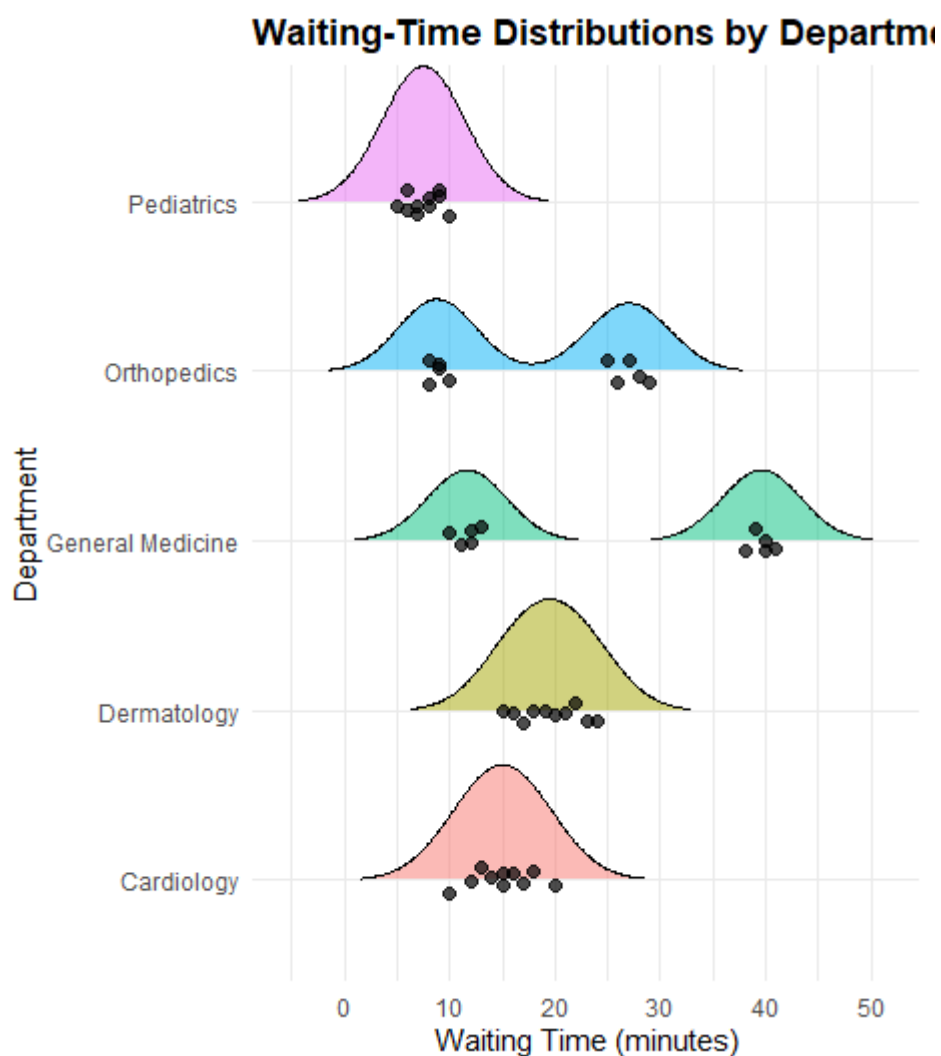
General Medicine

General Medicine is strongly bimodal:

- Approximately 10–13 minutes
- Approximately 38–41 minutes

This is the most extreme example of multimodality.

The distribution suggests that patients may be experiencing two fundamentally different waiting-time processes.



4. Variability Ranking

Using standard deviation:

Rank	Department	SD approximately
------	------------	------------------

1	General Medicine	14.80
---	------------------	-------

2	Orthopedics	9.67
---	-------------	------

3	Dermatology	3.03
---	-------------	------

4	Cardiology	2.94
---	------------	------

5	Pediatrics	1.58
---	------------	------

Therefore:

General Medicine > Orthopedics > Dermatology > Cardiology > Pediatrics

5. Median Waiting-Time Ranking

Rank	Department	Median
------	------------	--------

1	General Medicine	25.5
---	------------------	------

2	Dermatology	19.5
---	-------------	------

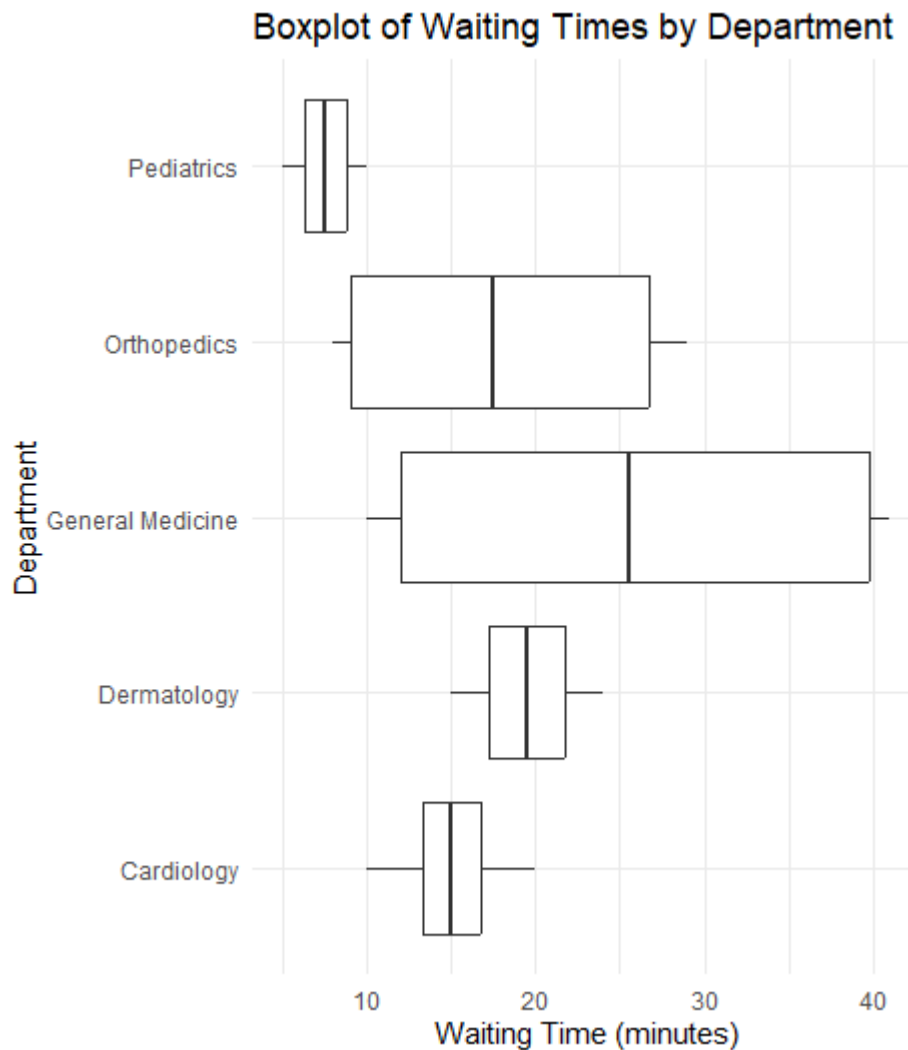
3	Orthopedics	17.5
---	-------------	------

4	Cardiology	15
---	------------	----

5	Pediatrics	7.5
---	------------	-----

Therefore:

General Medicine > Dermatology > Orthopedics > Cardiology > Pediatrics



6. Why the Rankings Differ

Variability and median measure different characteristics.

The **median** measures the central location of waiting times, whereas **standard deviation** measures how widely observations are dispersed.

General Medicine has both a high median and extremely high variability because its observations are split between very short and very long waiting times.

Orthopedics has a lower median than Dermatology but substantially greater variability because it contains two separated clusters.

7. Comparison with Boxplot

A boxplot summarizes:

- Median
- Quartiles
- Interquartile range
- Potential outliers

However, a standard boxplot can conceal the **shape of the distribution**.

In particular, the boxplot may fail to clearly show:

- Bimodality
- Multiple clusters
- Gaps
- Density concentration
- Separate patient populations

The bean/density visualization provides more information about the underlying distribution.

8. Staffing Recommendation

The strongest staffing intervention should be made in **General Medicine**.

The distribution contains two very different waiting-time groups. A possible explanation is insufficient staffing during certain periods or a separate category of complex patients.

Recommendation

Introduce **peak-hour staffing or a separate scheduling queue for complex cases** in General Medicine.

Orthopedics should also be investigated because of its bimodal waiting-time pattern.

Final conclusion

Pediatrics has the most consistent waiting times, while General Medicine has the greatest variability and strongest bimodality. General Medicine should receive priority for staffing and scheduling intervention.

QUESTION 3 — Choosing an Honest Proportion Chart

1. Objective

The objective is to compare payment-method preferences across three store formats using grouped bars, 100% stacked bars and pie charts while identifying visualization choices that may create misleading impressions.

2. R Code

```
# =====  
# QUESTION 3 - PROPORTION VISUALIZATION  
# =====  
  
library(ggplot2)  
  
# -----  
# Create dataset  
# -----  
  
payment_data <- data.frame(  
  Store_Format = c(  
    "Hypermarket",  
    "Supermarket",  
    "Convenience"  
  ),  
  Cash = c(18, 22, 35),  
  Card = c(34, 30, 25),  
  UPI_Wallet = c(48, 48, 40)  
)  
  
# Convert to long format  
payment_long <- data.frame(  
  Store_Format = rep(payment_data$Store_Format, each = 3),
```

```

Payment_Method = rep(
  c("Cash", "Card", "UPI/Wallet"),
  times = 3
),
Percentage = c(
  18, 34, 48,
  22, 30, 48,
  35, 25, 40
)
)

# =====

# TASK 1: GROUPED BAR CHART

# =====

grouped_bar <- ggplot(
  payment_long,
  aes(
    x = Store_Format,
    y = Percentage,
    fill = Payment_Method
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7
  ) +
  labs(

```

```

    title = "Payment Method by Store Format",
    x = "Store Format",
    y = "Percentage of Customers"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold")
  )

print(grouped_bar)

# =====

# TASK 2: 100% STACKED BAR CHART

# =====

stacked_bar <- ggplot(
  payment_long,
  aes(
    x = Store_Format,
    y = Percentage,
    fill = Payment_Method
  )
) +
  geom_col(position = "fill") +
  scale_y_continuous(
    labels = scales::percent
  ) +
  labs(

```



```

    title = "Payment Method Composition by Store Format",
    x = "Store Format",
    y = "Percentage"
  ) +
  theme_minimal()

print(stacked_bar)

# =====

# TASK 3: THREE PIE CHARTS

# =====

pie_plot <- ggplot(
  payment_long,
  aes(
    x = "",
    y = Percentage,
    fill = Payment_Method
  )
) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  facet_wrap(~ Store_Format) +
  labs(
    title = "Payment Method Distribution by Store Format"
  ) +
  theme_void() +
  theme(

```

```

    plot.title = element_text(face = "bold")
)

print(pie_plot)

# =====
# Print the data
# =====

print(payment_data)

```

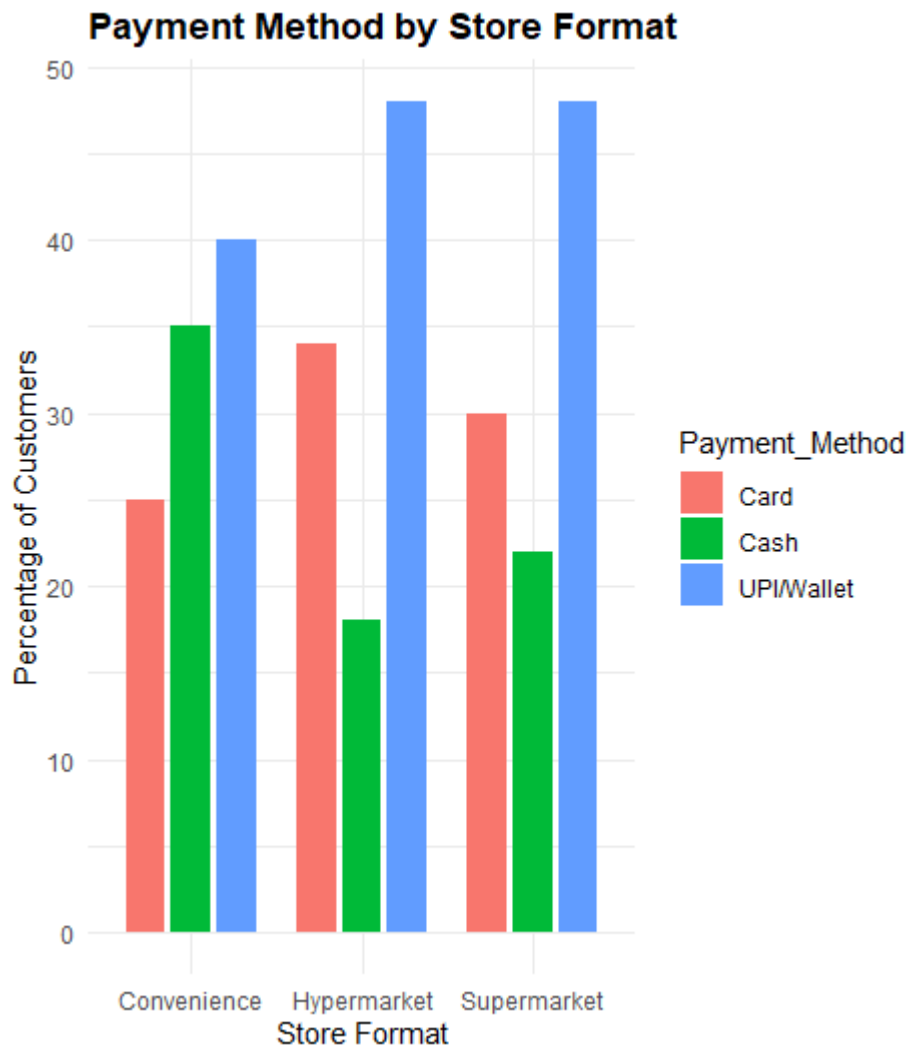
3. Grouped Bar Chart Interpretation

The grouped bar chart allows direct comparison of the three payment methods across formats.

UPI/Wallet share:

- Hypermarket = **48%**
- Supermarket = **48%**
- Convenience = **40%**

Therefore, **Hypermarket and Supermarket have the highest UPI/Wallet share**, tied at 48%.



4. 100% Stacked Bar Interpretation

The 100% stacked bar chart clearly shows the composition within each store format.

Hypermarket

- Cash = 18%
- Card = 34%
- UPI/Wallet = 48%

Supermarket

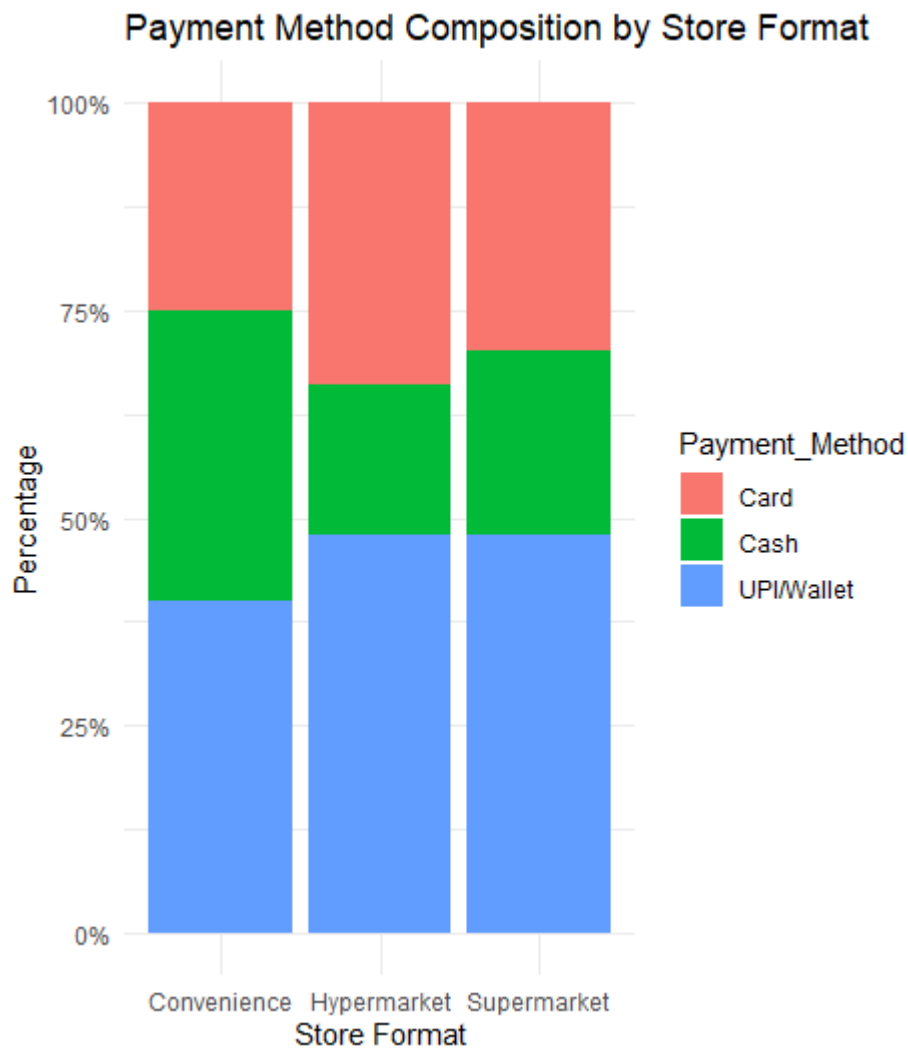
- Cash = 22%
- Card = 30%
- UPI/Wallet = 48%

Convenience

- Cash = 35%
- Card = 25%
- UPI/Wallet = 40%

The stacked chart is useful for showing the **overall composition** of each store format.

However, comparing internal segments can be difficult because not all segments share the same baseline.



5. Pie Chart Comparison

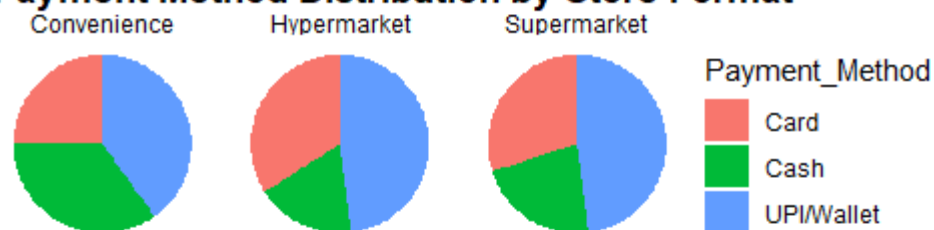
Pie charts represent each payment method using angular sectors.

The viewer can identify that UPI/Wallet occupies the largest sector in all three formats.

However, comparing **48% versus 40%** across separate pie charts is more difficult than comparing bars with a common baseline.

Therefore, the grouped bar chart is the most accurate visualization for comparing UPI/Wallet shares.

Payment Method Distribution by Store Format



6. Arc-Length Versus Length Perception

Bar charts use **length from a common baseline**, making small differences easier to judge.

Pie charts use **angles and arc lengths**.

Humans are generally better at comparing aligned lengths than comparing angles.

For example:

[
48% - 40% = 8%
]

The 8-percentage-point difference is immediately visible in a grouped bar chart but less precise when comparing two separate pie slices.

Conclusion

For management dashboards requiring accurate comparison, the **grouped bar chart is preferred**.

7. Misleading Visualization Example

One misleading technique would be to **truncate the y-axis**.

For example, if the vertical axis starts at 30% instead of 0%, Convenience's cash share of 35% could appear dramatically larger than Hypermarket's 18%.

This exaggerates the visual difference.

Another misleading technique would be a **3D pie chart**, because perspective can make slices closer to the viewer appear larger than they actually are.

Honest design recommendation

Use:

- A zero-based y-axis
- 2D bars
- Consistent category ordering
- Clear labels
- No unnecessary 3D effects

Final conclusion

The grouped bar chart is the best choice for accurate comparison of payment-method shares. Hypermarket and Supermarket have the highest UPI/Wallet share at 48%, while Convenience has the highest cash share at 35%.

QUESTION 4 — Nested Proportions in a Household Budget

1. Objective

The objective is to visualize a ₹60,000 monthly household budget at both category and subcategory levels using a treemap and sunburst chart.

2. R Code — Treemap

```
# =====
```

```
# QUESTION 4 - TREEMAP AND SUNBURST
```

```
# =====
```

```
# Install packages if required:
```

```
# install.packages("treemapify")
```

```
# install.packages("plotly")
```

```
library(ggplot2)
```

```
library(treemapify)
```

```
library(plotly)
```

```
# -----
```

```
# Create dataset
```

```
# -----
```

```
budget_data <- data.frame(
```

```
  Category = c(
```

```
    "Housing", "Housing", "Housing",
```

```
    "Food", "Food",
```

```
    "Transport", "Transport",
```

```
    "Savings", "Savings",
```

```
    "Discretionary", "Discretionary"
```

```
  ),
```

```
  Subcategory = c(
```

```
    "Rent/EMI",
```

"Utilities",
"Maintenance",
"Groceries",
"Dining Out",
"Fuel",
"Public Transit",
"Investments",
"Emergency Fund",
"Entertainment",
"Shopping"

),

Amount = c(

18000,

4000,

2000,

9000,

3000,

3500,

1500,

10000,

4000,

3000,

2000

)

)

=====


```
# TASK 1: TREEMAP
```

```
# =====
```

```
treemap_plot <- ggplot(  
  budget_data,  
  aes(  
    area = Amount,  
    fill = Category,  
    label = paste0(  
      Subcategory,  
      "\n₹",  
      format(Amount, big.mark = ";")  
    )  
  )  
) +  
  geom_treemap() +  
  geom_treemap_text(  
    colour = "white",  
    place = "centre",  
    grow = TRUE  
  ) +  
  labs(  
    title = "Household Monthly Budget Treemap"  
  ) +  
  theme_minimal()  
  
print(treemap_plot)
```

```

# =====

# TASK 2: SUNBURST

# =====

fig <- plot_ly(
  budget_data,
  ids = ~Subcategory,
  labels = ~Subcategory,
  parents = ~Category,
  values = ~Amount,
  type = "sunburst",
  branchvalues = "total"
)

fig <- fig %>%
  layout(
    title = "Household Budget Sunburst Chart"
  )

fig

```

3. Category-Level Calculation

The total budget is:

```
[
₹60,000
]
```

Housing consists of:

$$[18,000 + 4,000 + 2,000]$$

$$[= ₹24,000]$$

Therefore:

$$[\text{Housing\ Percentage} = \frac{24,000}{60,000} \times 100]$$

$$[=40\%]$$

Housing consumes 40% of the total budget.

4. Largest Housing Subcategory

The Housing subcategories are:

Subcategory Amount

Rent/EMI ₹18,000

Utilities ₹4,000

Maintenance ₹2,000

The largest is **Rent/EMI = ₹18,000.**

Percentage of total budget:

$$[\frac{18,000}{60,000} \times 100 = 30\%]$$

Therefore:

- Housing = **40%**
- Rent/EMI = **30% of total budget**

- Rent/EMI = **75% of Housing spending**

because:

$$\left[\frac{18,000}{24,000} \times 100 = 75\% \right]$$

5. Complete Budget Summary

Category	Amount	Percentage
Housing	₹24,000	40.00%
Food	₹12,000	20.00%
Transport	₹5,000	8.33%
Savings	₹14,000	23.33%
Discretionary	₹5,000	8.33%
Total	₹60,000	100%

6. Treemap Versus Sunburst

Treemap

The treemap uses **area** to represent monetary value.

Advantages:

- Efficient use of screen space
- Easy to see large categories
- Shows category and subcategory hierarchy
- Good for dashboards

Disadvantage:

Comparing two rectangles from different parts of the treemap can be difficult because their shapes and positions differ.

Sunburst

The sunburst uses concentric rings.

- Inner ring = major categories
- Outer ring = subcategories

Advantages:

- Clearly communicates hierarchy
- Easy to understand parent-child relationships
- Good for showing nested structure

Disadvantage:

Comparing the exact size of two outer-ring segments is harder because the segments are curved.

7. Which Is Better for Cross-Category Comparison?

Suppose we want to compare:

- Food → Groceries = ₹9,000
- Savings → Emergency Fund = ₹4,000

The treemap provides direct area comparison, but different rectangle shapes can still make precise comparison difficult.

The sunburst makes the hierarchy very clear but makes exact comparison of different outer-ring segments even harder because they are represented by arcs.

Therefore, if **accurate numerical comparison** is the main objective, a bar chart would actually be better than either visualization.

For showing **hierarchy**, the treemap and sunburst are appropriate.

8. Treemap Misleading-Design Analysis

A treemap can cause misjudgment because viewers estimate area imperfectly.

For example, two categories could have similar areas but very different shapes:

- One may be a long, thin rectangle.
- Another may be a compact square.

The viewer may perceive the compact square as more substantial even when the areas are equal.

Therefore, the treemap should be accompanied by:

- Numerical labels
- Consistent scaling
- Clear category labels

Final conclusion

Housing consumes 40% of the monthly budget, with Rent/EMI alone consuming 30% of the total budget. The treemap is effective for area-based hierarchy, while the sunburst is effective for showing parent-child relationships.

QUESTION 5 — Flow Analysis Using Sankey Diagram and Parallel Sets

1. Objective

The objective is to analyze the flow of 250 students from course enrollment through device usage to final completion status.

2. R Code — Sankey Diagram

```
# =====  
  
# QUESTION 5 - SANKEY DIAGRAM AND PARALLEL SETS  
  
# =====  
  
# Install packages if required:  
# install.packages("networkD3")  
# install.packages("ggalluvial")  
# install.packages("ggplot2")  
# install.packages("dplyr")  
  
library(networkD3)  
library(ggalluvial)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
# -----
```

```
# Create dataset
```

```
# -----
```

```
student_data <- data.frame(
```

```
  Course = c(
```

```
    "Data Science", "Data Science",
```

```
    "Data Science", "Data Science",
```

```
    "Web Development", "Web Development",
```

```
    "Web Development", "Web Development",
```

```
    "Digital Marketing", "Digital Marketing",
```

```
    "Digital Marketing", "Digital Marketing"
```

```
  ),
```

```
  Device = c(
```

```
    "Mobile", "Mobile",
```

```
    "Laptop", "Laptop",
```

```
    "Mobile", "Mobile",
```

```
    "Laptop", "Laptop",
```

```
    "Mobile", "Mobile",
```

```
    "Laptop", "Laptop"
```

```
  ),
```

```
  Status = c(
```

```
    "Completed", "Dropped",
```

```
"Completed", "Dropped",  
"Completed", "Dropped",  
"Completed", "Dropped",  
"Completed", "Dropped",  
"Completed", "Dropped"  
)
```

```
Students = c(  
  20, 25,  
  45, 10,  
  15, 20,  
  35, 10,  
  10, 15,  
  20, 25  
)  
)
```

```
print(student_data)
```

```
# =====
```

```
# TASK 1: SANKEY DIAGRAM
```

```
# =====
```

```
# Create nodes
```

```
nodes <- data.frame(  
  name = unique(c(  
    as.character(student_data$Course),  
    as.character(student_data$Device),
```



```
    as.character(student_data$Status)
  ))
)
```

```
# Course -> Device links
```

```
course_device <- student_data %>%
  group_by(Course, Device) %>%
  summarise(
    Students = sum(Students),
    .groups = "drop"
  ) %>%
  rename(
    Source = Course,
    Target = Device
  )
```

```
# Device -> Status links
```

```
device_status <- student_data %>%
  group_by(Device, Status) %>%
  summarise(
    Students = sum(Students),
    .groups = "drop"
  ) %>%
  rename(
    Source = Device,
    Target = Status
  )
```

```
links <- bind_rows(  
  course_device,  
  device_status  
)
```

```
links$source <- match(  
  links$Source,  
  nodes$name  
) - 1
```

```
links$target <- match(  
  links$Target,  
  nodes$name  
) - 1
```

```
links$value <- links$Students
```

```
# Create Sankey diagram  
sankey <- sankeyNetwork(  
  Links = links,  
  Nodes = nodes,  
  Source = "source",  
  Target = "target",  
  Value = "value",  
  NodeID = "name",  
  fontSize = 13,  
  nodeWidth = 30,  
  sinksRight = FALSE
```

)

sankey

3. R Code — Parallel Sets / Alluvial Diagram

```
# =====  
# PARALLEL SETS / ALLUVIAL DIAGRAM  
# =====
```

```
alluvial_plot <- ggplot(  
  student_data,  
  aes(  
    axis1 = Course,  
    axis2 = Device,  
    axis3 = Status,  
    y = Students  
  )  
) +  
  geom_alluvium(  
    aes(fill = Status),  
    alpha = 0.7  
  ) +  
  geom_stratum(  
    width = 0.25,  
    fill = "grey80",  
    colour = "black"  
  ) +  
  geom_text(  
    aes(x = 1, y = 1, text = "Alluvial Diagram")  
  )
```

```

stat = "stratum",
aes(label = after_stat(stratum)),
size = 3
) +
scale_x_discrete(
  limits = c(
    "Course",
    "Device",
    "Completion Status"
  ),
  expand = c(0.1, 0.1)
) +
labs(
  title = "Student Flow: Course → Device → Completion Status",
  x = "",
  y = "Number of Students"
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold")
)

print(alluvial_plot)

```

4. Calculate the Three Largest Flows

The individual Course → Device → Completion Status flows are:

Course	Device	Status	Students	Percentage
Data Science	Laptop	Completed	45	18%
Web Development	Laptop	Completed	35	14%
Data Science	Mobile	Dropped	25	10%
Digital Marketing	Laptop	Dropped	25	10%
Data Science	Mobile	Completed	20	8%
Web Development	Mobile	Dropped	20	8%
Digital Marketing	Laptop	Completed	20	8%
Web Development	Mobile	Completed	15	6%
Digital Marketing	Mobile	Dropped	15	6%
Data Science	Laptop	Dropped	10	4%
Web Development	Laptop	Dropped	10	4%
Digital Marketing	Mobile	Completed	10	4%

The three largest flows are:

1. Data Science → Laptop → Completed

$$\left[\frac{45}{250} \times 100 = 18\% \right]$$

2. Web Development → Laptop → Completed

$$\left[\frac{35}{250} \times 100 = 14\% \right]$$

3. Data Science → Mobile → Dropped

$$\left[\frac{25}{250} \times 100 = 10\% \right]$$

There is also a tie for the third-largest value because:

Digital Marketing → Laptop → Dropped = 25 students = 10%.

Therefore, the dataset technically has **four flows involved in the top-three threshold** because two flows are tied at 25 students.

5. Course Completion Analysis

Data Science

Total:

$$\begin{bmatrix} 20+25+45+10=100 \\ \end{bmatrix}$$

Completed:

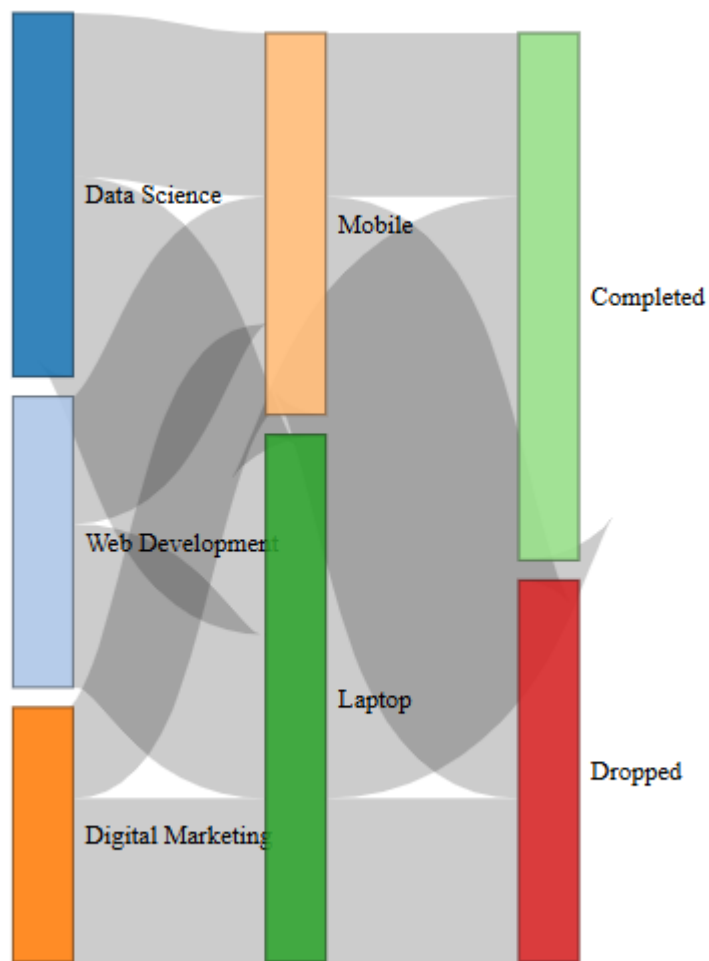
$$\begin{bmatrix} 20+45=65 \\ \end{bmatrix}$$

Completion rate:

$$\begin{bmatrix} \frac{65}{100} \times 100 = 65\% \\ \end{bmatrix}$$

Dropout rate:

$$\begin{bmatrix} 35\% \\ \end{bmatrix}$$



Web Development

Total:

$$[15+20+35+10=80]$$

Completed:

$$[15+35=50]$$

Completion rate:

$$\left[\frac{50}{80} \times 100 = 62.5\% \right]$$

Dropout rate:

$$\left[37.5\% \right]$$

Digital Marketing

Total:

$$\left[10 + 15 + 20 + 25 = 70 \right]$$

Completed:

$$\left[10 + 20 = 30 \right]$$

Completion rate:

$$\left[\frac{30}{70} \times 100 \approx 42.86\% \right]$$

Dropout rate:

$$\left[57.14\% \right]$$

6. Course with Weakest Completion Outcome

Course	Students Completed		Completion Rate	Dropout Rate
Data Science	100	65	65.00%	35.00%
Web Development	80	50	62.50%	37.50%

Course	Students	Completed	Completion Rate	Dropout Rate
Digital Marketing	70	30	42.86%	57.14%

Therefore, **Digital Marketing has the weakest completion outcome.**

Its dropout rate is approximately:

[
57.14%
]

This means that more than half of the Digital Marketing students dropped out.

7. Where Is the Drop-Off Most Visible?

For Digital Marketing:

Mobile

- Completed = 10
- Dropped = 15

Dropout rate:

[
 $\frac{15}{25} \times 100 = 60\%$
]

Laptop

- Completed = 20
- Dropped = 25

Dropout rate:

[
 $\frac{25}{45} \times 100 \approx 55.56\%$
]

The mobile pathway has a slightly higher dropout rate.

However, the **largest absolute Digital Marketing dropout flow** is:

[
Digital \ Marketing $\rightarrow$ Laptop $\rightarrow$ Dropped
]

with:

[
25\ students = 10%
]

of the total dataset.

Therefore, the Sankey diagram should show a particularly noticeable flow from **Digital Marketing → Laptop → Dropped**.

8. Misleading Sankey / Parallel-Set Design

One major source of misleading information is using **unscaled link widths**.

In a proper Sankey diagram:

[
Link\ Width \propto Flow\ Size
]

For example:

- 45 students should have a link 4.5 times as large as a 10-student flow.
- 25 students should be visually larger than 10 students.

If every link is drawn with the same width, a viewer could incorrectly assume that all pathways have similar importance.

Another problem is inconsistent node ordering.

For example, if "Completed" appears in different positions for different courses, visual comparison becomes difficult.

Best practice

Use:

- Consistent node ordering
 - Proportional link widths
 - Clear labels
 - Consistent colors for completion status
 - Numerical annotations where appropriate
-

9. Overall Conclusion for Question 5

The Sankey and parallel-set visualizations show that **Data Science has the strongest completion performance**, while **Digital Marketing has the weakest**.

Digital Marketing has only **42.86% completion** and **57.14% dropout**.

The largest individual pathway is:

Data Science → Laptop → Completed = 45 students (18%)

followed by:

Web Development → Laptop → Completed = 35 students (14%)

The major concern is the high dropout rate in Digital Marketing, especially the laptop pathway with **25 dropped students**.

OVERALL PROJECT CONCLUSION

The five visualization techniques provide different perspectives on the datasets.

Question 1 — ECDF and Q-Q Plots

Partner B provides more consistent delivery performance. Partner A has greater variability and a long right tail caused by extreme delivery times. Percentile-based SLAs are preferable to $\text{mean} \pm \text{standard deviation}$.

Question 2 — Bean Plots

Pediatrics has the lowest variability, while General Medicine has the highest variability. Orthopedics and General Medicine display bimodal distributions, suggesting the presence of different operational processes or patient groups.

Question 3 — Proportions

Grouped bar charts provide the most accurate comparison of payment-method shares because they use a common baseline. Pie charts are less precise for comparing small differences. UPI/Wallet has the highest share in Hypermarket and Supermarket at 48%.

Question 4 — Nested Proportions

Housing consumes **40%** of the total ₹60,000 budget. Rent/EMI alone accounts for **₹18,000 or 30% of the total budget**. Treemaps are useful for area-based hierarchical visualization, while sunbursts clearly communicate nested structure.

Question 5 — Flow Analysis

Data Science has the highest completion rate at **65%**, Web Development has **62.5%**, and Digital Marketing has the lowest at **42.86%**. Sankey and parallel-set diagrams effectively reveal dominant pathways and dropout patterns.